ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C7 · LIGHT AND SEEING

Light sources, mirrors and the eye

★☆☆
Grade 5

Learning goals

- Identify light sources.
- Label the eye.
- Recall what optical devices do.

1 Sort these objects. [8]

SORT THESE

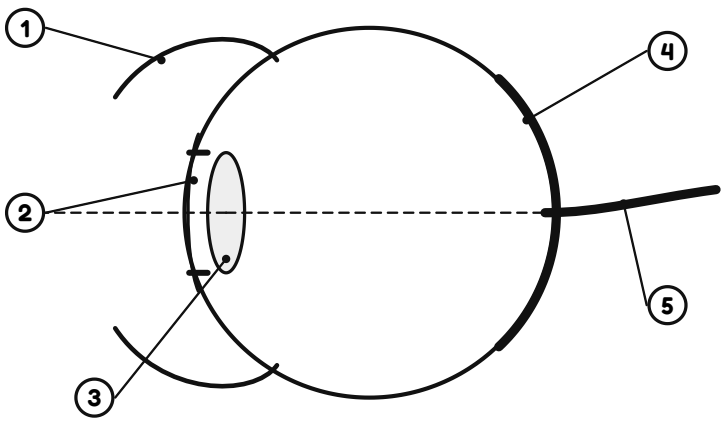
a white page **an LED bulb** **a candle flame** **the Sun** **a shiny spoon** **a firefly** **a mirror** **the Moon**

Light source	Not a light source (reflects light)
the Sun	the Moon
a candle flame	a mirror
a firefly	a white page
an LED bulb	a shiny spoon

2 Label the parts of the eye. [5]

LABELS

eyelid **pupil** **retina** **lens** **optic nerve**



1 eyelid
2 pupil
3 lens
4 retina
5 optic nerve

3 Match each optical device to what it does. [5]

Draw a line to match each pair, or write the letter.

hand lens ●	→ E	● **A**	makes an upside-down image through a tiny hole
telescope ●	→ D	● **B**	uses mirrors to make repeating patterns
pinhole camera ●	→ A	● **C**	uses two mirrors to see over a wall
kaleidoscope ●	→ B	● **D**	makes far-away objects look closer
periscope ●	→ C	● **E**	makes small things look bigger

4 True or false? [4]

Statement	True	False	If false, correct it
1. The Moon is a light source.		✔	The Moon reflects light from the Sun.
2. Light hits a mirror and bounces off at the same angle.	✔		
3. Our eyes send out rays so that we can see.		✔	Light enters our eyes from sources or reflected from objects.
4. The pupil gets bigger in the dark.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C7 · LIGHT AND SEEING

Tracing the path of light

★★☆
Grade 5

Learning goals

- Draw ray diagrams.
- Explain reflection, refraction and lateral inversion.

1 **Draw arrows to show how Mina sees the mango. Label the light source.** [2]

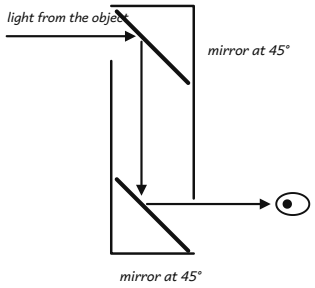
Sun → mango → Mina's eye

An arrow from the Sun to the mango, then an arrow from the mango into Mina's eye. The Sun is labelled as the light source.

2 **A ray of light hits a mirror at an angle of 35° to the normal. What is the angle of the reflected ray? Explain.** [2]

35°: light is reflected at the same angle as the incoming ray.

3 **Look at the periscope.** [3]



a) Why must both mirrors be at 45°?

So the light turns through a right angle at each mirror: first down the tube, then out to the eye.

b) Give one real use of a periscope.

seeing over a crowd or wall; seeing above water from a submarine; seeing round corners

4 **Rafi writes his name on a card and holds it up to a mirror. Describe what he sees and use the scientific term.** [2]

The letters are back to front (reversed left to right). This is lateral inversion.

5 **A fisherman spears a fish that looks close to the water surface, but he misses. Use refraction to explain why.** [2]

Light from the fish bends as it leaves the water, so the fish looks higher (shallower) than it really is. He must aim lower.

6 **Why does your pupil get smaller when you walk out into bright sunshine?** [1]

To let less light into the eye and protect the retina from too much light.

ANSWER KEY

INVESTIGATE · C7 · LIGHT AND SEEING

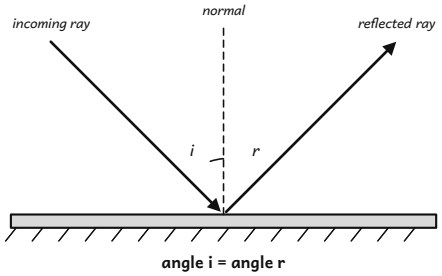
Investigating mirrors: angles of reflection

★★☆ Grade 5

Learning goals

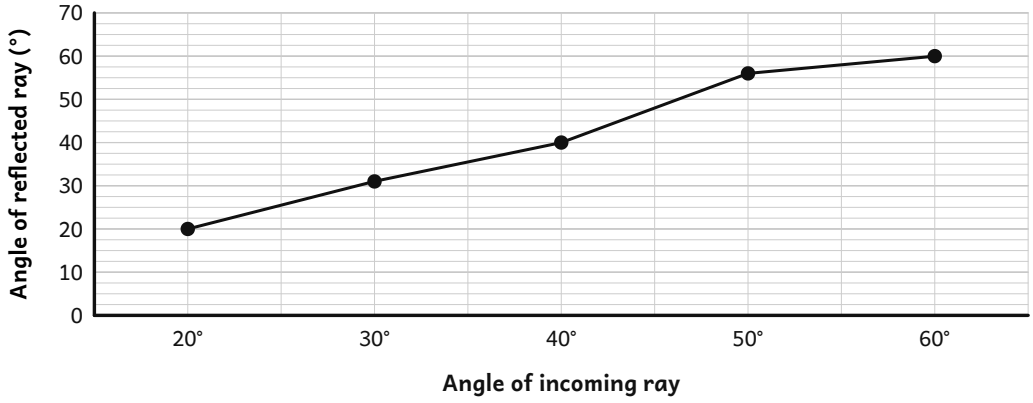
- Measure angles with a protractor.
- Spot a result that does not fit.
- Draw a conclusion.

Class 5C shone a thin beam of light from a ray box onto a plane mirror. They drew the normal, then used a protractor to measure the angle of the incoming ray and the angle of the reflected ray.



1 **Here are the results.** [5]

Angle of incoming ray (°)	20	30	40	50	60
Angle of reflected ray (°)	20	31	40	56	60



a) Plot the results as a line graph.

b) Which result does **not** fit the pattern?

At 50° the reflected ray was measured as 56°.

c) Suggest a reason for this result and what the class should do.

They may have misread the protractor or drawn the normal wrongly; they should repeat the measurement.

2 **Write a conclusion.** [2]

The angle of the reflected ray is equal to the angle of the incoming ray.

3 **Predict the angle of reflection for an incoming ray at 75°.** [1]

75°

4 **Why did they use a thin beam of light rather than a wide torch beam?** [1]

A thin beam is a single clear ray, so the angles can be measured accurately.

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C7 · LIGHT AND SEEING

Healthy eyes in a screen world

★★★
Grade 5

Learning goals

- Use data from a survey.
- Link eye health to the structure of the eye.
- Plan a health campaign.

Science | Health | Maths | Technology | Wellbeing

More and more children around the world are becoming **short-sighted** (myopic): they see near things clearly, but far things look blurred, because the image forms in front of the retina instead of on it. Scientists estimate that by 2050 about **half of the world's people** could be short-sighted.

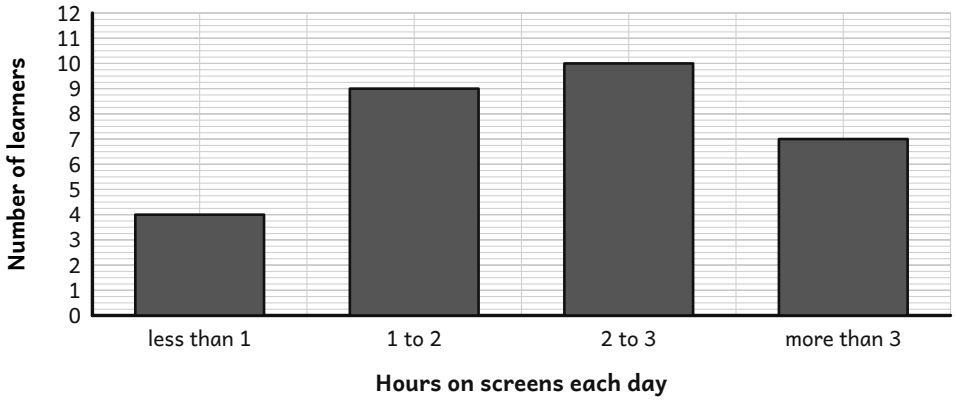
Research suggests that spending time **outdoors in daylight** helps to protect children's eyes. In Taiwan, schools encouraged children to spend 120 minutes outdoors every day, and the number of short-sighted children began to fall. Eye doctors also recommend the **20-20-20 rule**: every 20 minutes of close work or screen use, look at something about 6 metres (20 feet) away for 20 seconds.

Food matters too. Without enough **vitamin A**, people can develop **night blindness**. Bangladesh runs a national Vitamin A campaign that gives capsules to young children, and vitamin A comes from foods such as spinach, carrots, mangoes, eggs and small fish like *mola*.

Based on research reported in the journal Ophthalmology (2016) and Taiwan school studies; Bangladesh National Vitamin A Plus Campaign.

1 **Class 5C did a survey of 30 learners: "How many hours do you spend on screens each day, outside school?" (example data)** [3]

Hours per day	less than 1	1 to 2	2 to 3	more than 3
Number of learners	4	9	10	7



a) Draw the bar chart.

b) How many learners spend 2 hours or more on screens each day?

17 learners (10 + 7)

2 **Use your knowledge of the eye to explain why a short-sighted person sees far objects as blurred.** [2]

The lens focuses the light in front of the retina instead of on it, so the image on the retina is not sharp.

3 **What might each person say about screens?** [3]

Person	View
a learner	Screens help me learn and talk to my cousins, but my eyes get tired.
a parent	I worry about too much screen time and too little outdoor play.
an eye doctor	Take breaks, follow 20-20-20, play outside every day and get eyes checked.

4 **Design a "Healthy Eyes Week" for your school. Give three actions and the reason for each.** [3]

e.g. outdoor play every break (daylight protects eyes); 20-20-20 reminders in computer lessons (rests the eye muscles); vitamin A food stall (prevents night blindness); free eye tests (find problems early).

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C7 · LIGHT AND SEEING

Show what you know

★★★
Grade 5

Learning goals

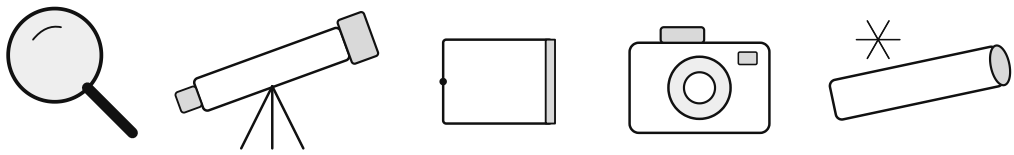
- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

L	C	**R**	**E**	**T**	**I**	**N**	**A**	L	P	U
F	O	D	E	R	N	**S**	J	A	Y	R
U	U	**M**	**I**	**R**	**R**	**O**	**R**	A	R	T
F	H	U	G	O	**E**	**U**	J	E	F	M
D	C	N	C	A	**F**	**R**	T	J	J	O
Y	W	H	M	M	**L**	**C**	Y	E	C	V
D	V	Y	T	K	**E**	**E**	A	**P**	**L**	**N**
B	H	G	L	W	**C**	C	K	**U**	**I**	**E**
B	**L**	**E**	**N**	**S**	**T**	A	B	**P**	**G**	**R**
I	**M**	**A**	**G**	**E**	F	M	M	**I**	**H**	**V**
Y	O	A	H	P	M	M	E	**L**	**T**	**E**

image **lens** **light** **mirror** **nerve** **pupil** **reflect** **retina** **source**

2 Write the letter of each optical device. [5]



A B C D E

Device	Letter
camera	D
kaleidoscope	E
hand lens	A
pinhole camera	C
telescope	B

3 Which part of the eye has light-sensitive cells? [1]

○ eyelid
○ pupil
● **retina**
○ lens

4 **A straw in a glass of water looks bent. This is because of ...** [1]

◯ reflection
● **refraction**
◯ absorption
◯ lateral inversion

5 **Challenge. Explain how you can see this page. Use the words *source*, *reflect*, *pupil*, *retina*, *nerve* and *brain*.** [3]

Light from a source (the Sun or a lamp) reflects off the page and enters my eye through the pupil. The lens focuses it on the retina, which sends messages along the optic nerve to my brain.